



Universidad de las Fuerzas Armadas - ESPE

Object Oriented Programming

NRC: 30812

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Requirements and Features

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Requirements and Features

1. **User Registration:** The system must allow the creation of profiles for mothers and babies using the Profile Manager function. Required fields include: First Name, Last Name, Weight, Height, Date of Birth, National Identification Number (10-digit validation), and Disability Status.
2. **Pregnancy Tracking:** The system must calculate the current week of gestation using the Pregnancy Calculator function, based on the date of the last menstrual period (LMP).
3. **Growth Status Validation:** The system will compare the baby's anthropometric data (weight/height) with normal standards for their age, issuing a status message: Good/Normal, Careful, or Critical.
4. **Medical History:** The system must record appointments and vital signs using the Medical Check-up Record function.
5. **Warning Signs:** The system will display critical alerts requiring immediate hospital attention:
 - Temperature: Fever ($>38^{\circ}\text{C}$) or hypothermia ($<35.5^{\circ}\text{C}$).
 - Respiratory distress: Rib retractions, nasal flaring, or grunting.
 - Feeding: Refusal to suckle or complete vomiting of ingested food.
 - Skin color: Cyanosis (purple), extreme pallor, or jaundice (yellowing) extending to the legs.
 - Activity level: Lethargy (difficult to awaken) or inconsolable crying.

6. Community interaction: The system should provide a space for user questions through an interactive chat feature.
7. Parenting tips: The system will offer a visual repository of tips organized by category. Each tip will be presented as a slideshow that guides the user
8. Search tool: The system should allow users to find specific tips by searching by keywords.
9. Schedule an appointment: Mothers will be able to schedule the dates and times for their medical consultation.